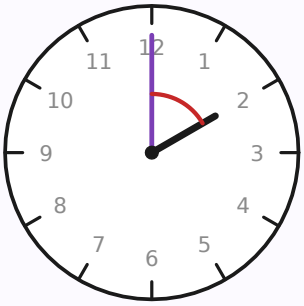
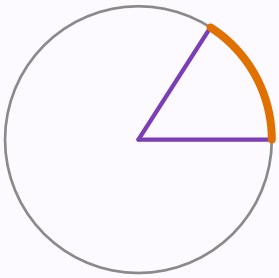
Chapter 27

Angles: degrees and radians

how much have you turned?



2 o'clock: the hands are 60° apart



string one radius long, laid on the rim: 1 radian

An angle isn't a length or a shape: it's **an amount of turning**. Open a door, turn a key, spin round on the spot: each is an angle.

We measure turning two ways: **degrees** (360 for a full turn, the everyday way) and **radians** (about 6.28 for a full turn, the way calculus and your lecturer prefer). This chapter shows where both come from and how to swap between them.

Goes with **Lecture 6** (radians vs degrees, notation and conversion).

∠ Turning on the spot

RULE: degrees and the names of angles

A full turn is $360°$. A half turn (a straight line) is $180°$. A quarter turn (a **right angle**) is $90°$.

Acute: less than $90°$. **Obtuse**: between $90°$ and $180°$. **Reflex**: more than $180°$.

Angles on a straight line add to $180°$; angles round a point add to $360°$.

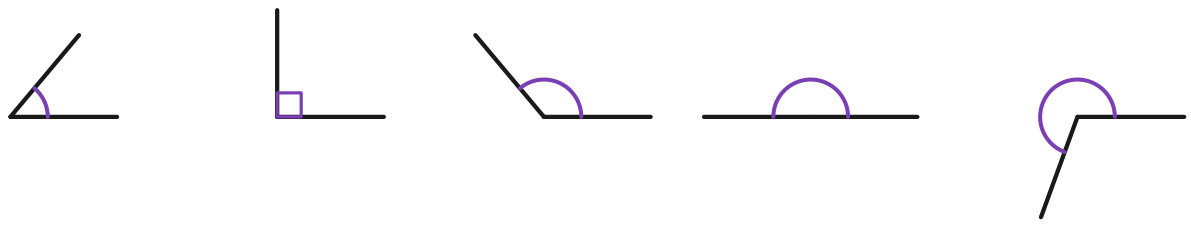
Stand facing north and turn. A **full turn** brings you back to where you started. We cut a full turn into **360 equal steps** called **degrees**, written with a little circle: 360°.



quarter turn **90°** | **half turn** **180°** | **three quarters** **270°** | **full turn** **360°**

Why 360? Nobody really knows: the Babylonians liked it (a year has about 360 days, and 360 divides by 2, 3, 4, 5, 6, 8, 9, 10, 12...). Lecture 6 calls degrees **arbitrary**, and that's fair.

Names for sizes of angle:



acute less than 90° | **right angle** exactly 90° | **obtuse** 90° to 180° | **straight** exactly 180° | **reflex** more than 180°

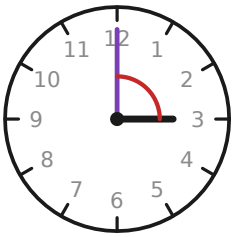
∠ Clock hands: angles you already read every day

RULE: angle between clock hands

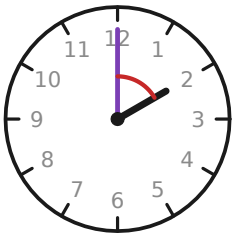
Measure both hands clockwise from 12. Minute hand: $6°$ per minute. Hour hand: $30°$ per hour **plus** $0.5°$ per minute past the hour.

The angle between them is the difference. If it is over $180°$, the smaller angle is $360°$ minus it.

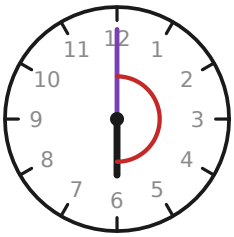
A clock face is a full turn split into 12 hours, so **each hour mark is 360 ÷ 12 = 30°** apart. The minute hand does a full turn in 60 minutes: **6° per minute**.



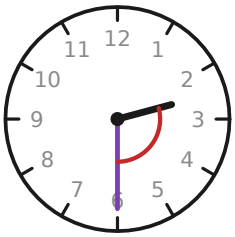
3:00
90°



2:00
60°



6:00
180°



2:30
105°

2 o'clock: the hour hand is 2 marks round, so $2 \times 30° = 60°$ (Lecture 6's cosine-rule clock, Chapter 31). The trap at 2:30: the hour hand has moved **halfway** towards 3, so it's at 75° from 12, while the minute hand is at 180°. Gap: $180° - 75° = 105°$, not 90°.

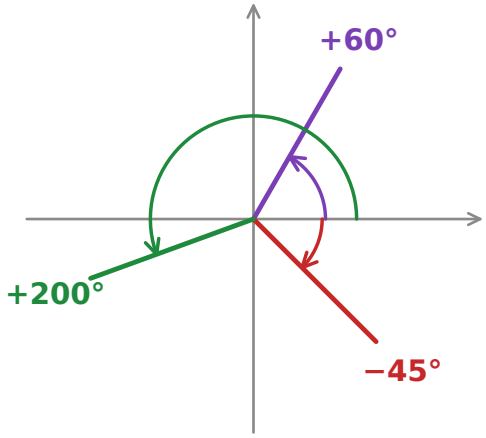
∠ Which way is positive? Anticlockwise, from the east

RULE: positive and negative angles

Maths angles are measured from the **positive x-axis** (east). **Anticlockwise is positive**, clockwise is negative.

Adding or subtracting $360°$ (2π) gives the **same direction**: $-45°$, $315°$ and $675°$ all point the same way.

Clocks go clockwise from 12. Mathematicians do the opposite: **start pointing east (along the positive x-axis) and turn anticlockwise**. Lecture 6 measures angles this way on the unit circle. Turning clockwise gives a **negative** angle.



+60°: a sixth of a turn anticlockwise
−45°: an eighth of a turn clockwise
+200°: past the half-turn

More than 360° just means more than one full turn: 550° = 360° + 190°. (Lecture 6 uses 550° in an example.)

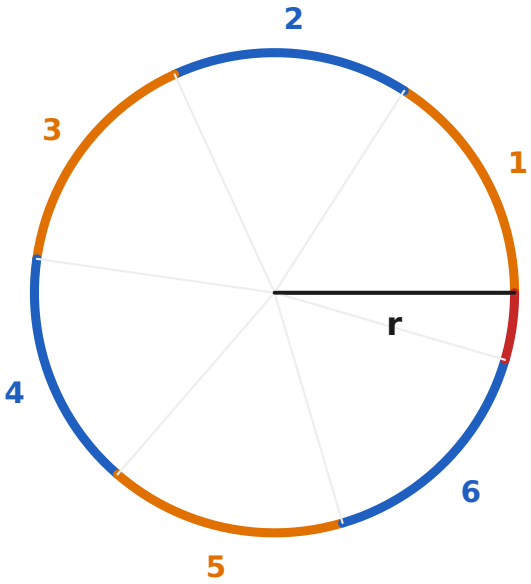
∠ The radian: a string one radius long

RULE: the radian

1 radian is the angle at the centre when the arc is exactly as long as the radius.

$2\pi \text{ rad} = 360°$, so $\pi \text{ rad} = 180°$ and $1 \text{ rad} \approx 57.3°$.

Take a piece of string **exactly as long as the circle's radius** and lay it along the rim. The angle it covers at the centre is **one radian**. (Lecture 6: "1 radian is the angle subtended by an arc of length R"; *subtended* just means "made at the centre by".)



Lay radius-long strings round the rim: 6 whole strings fit, plus a little red bit (0.28 of a string).

The circumference is 2πr (Chapter 5), so exactly 2π ≈ 6.28 strings fit.

So a full turn is 2π radians:

$$360° = 2\pi \text{ rad} \approx 6.28 \text{ rad}$$

$$1 \text{ rad} = \frac{360°}{2\pi} \approx 57.3°$$ ← *a bit less than 60°*

Why bother? A radian is **arc length ÷ radius**, a length divided by a length, so it has **no units**: it's a pure number. That makes formulas simpler (next pages) and calculus (Chapter 34) only works cleanly in radians.

∠ Converting: the unit-conversion chain again

RULE: converting and reducing angles

Degrees → radians: multiply by $\frac{\pi}{180}$. **Radians → degrees**: multiply by $\frac{180}{\pi}$.

Leave π in the answer when asked for an **exact** value ($60^\circ = \frac{\pi}{3}$); otherwise use $\pi \approx 3.1416$.

Reducing: an angle over 360° or below 0° points the same way as one between 0° and 360°. Subtract or add 360° (in radians, 2π) until it lands in range.

Chapter 5 converted units by multiplying by a fraction equal to 1 (like $\frac{100\text{ cm}}{1\text{ m}}$). Here the "1" is $\frac{\pi\text{ rad}}{180^\circ}$, because π rad and 180° are the same half-turn:

degrees → radians

$$\times \frac{\pi}{180}$$

radians → degrees

$$\times \frac{180}{\pi}$$

$$60^\circ \times \frac{\pi}{180} = \frac{60\pi}{180} = \frac{\pi}{3}$$ ← *cancel 60/180 = 1/3*

$$\frac{3\pi}{4} \times \frac{180}{\pi} = \frac{3 \times 180}{4} = 135^\circ$$ ← *the π's cancel*

Reducing angles into range (0° to 360°, or 0 to 2π):

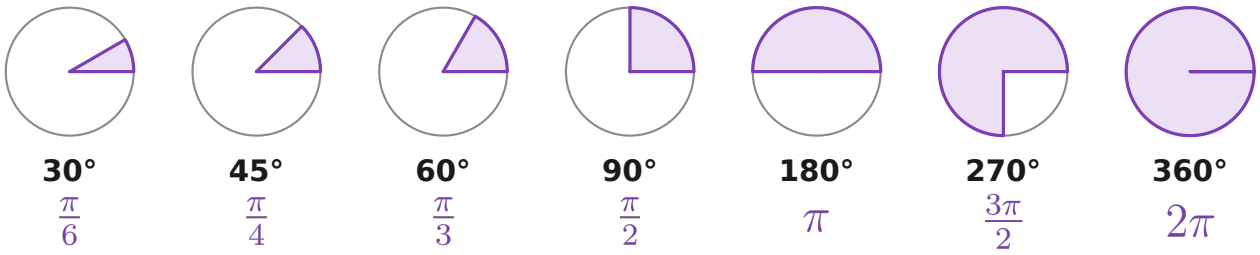
$$550^\circ - 360^\circ = 190^\circ$$ ← *over 360°: subtract a whole turn*

$$-45^\circ + 360^\circ = 315^\circ$$ ← *negative: add a whole turn*

$$\frac{9\pi}{4} - 2\pi = \frac{9\pi}{4} - \frac{8\pi}{4} = \frac{\pi}{4}$$ ← *in radians a whole turn is 2π*

$$-\frac{\pi}{3} + 2\pi = \frac{5\pi}{3}$$

Pizza shortcut: think in fractions of a turn. 90° is a quarter turn, and a quarter of 2π is $\frac{\pi}{2}$. The ones to know by heart:



30°	45°	60°	90°	180°	270°	360°
$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π

Notation warning (Lecture 6): degrees always carry the ° sign. Radians are written "rad", a small c (2^c), or **nothing at all**. So an angle with no ° is probably radians, but many sources are lazy: always check which one you're in.

∡ The calculator trap: DEG or RAD?

RULE: calculator mode

Angle written with ° → **DEG** mode. Angle with "rad", a π, or no unit at all → **RAD** mode.

Check the mode **before** every trig calculation. The calculator never warns you.

Your calculator has a mode: **DEG** or **RAD**. Type the same keys in the wrong mode and you get a confident, wrong answer:

DEG

```
sin(30)
                0.5000
```

✓

RAD

```
sin(30)
               -0.9880
```

✗

In RAD mode, "30" means 30 radians (almost 5 full turns), so sin gives -0.988 instead of 0.5. If a trig answer looks mad, check the mode first.

∠ Pizza maths: crust length and slice area

RULE: arc length and sector area

A slice is a **fraction of the whole circle**: arc = fraction × circumference $2\pi r$; sector = fraction × area πr^2.

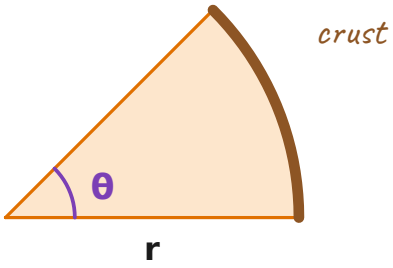
The fraction is $\dfrac{\theta}{360}$ when θ is in degrees, or $\dfrac{\theta}{2\pi}$ when θ is in radians.

That gives the two sets of formulas in the table.

Only use $s = r\theta$ and $A = \frac{1}{2}r^2\theta$ with θ in **radians**.

	θ in radians	θ in degrees
Arc length s	$s = r\theta$	$s = \dfrac{\theta}{360} \times 2\pi r$
Sector area A	$A = \frac{1}{2}r^2\theta$	$A = \dfrac{\theta}{360} \times \pi r^2$

With the angle θ in **radians**, the formulas for a slice of circle are beautifully simple (that's the payoff for radians):



crust

$$s = r\theta$$ *arc (crust) length*

$$A = \frac{1}{2}r^2\theta$$ *sector (slice) area*

Why they work: a full turn ($\theta = 2\pi$) gives $s = 2\pi r$ (the whole circumference) and $A = \frac{1}{2}r^2 \cdot 2\pi = \pi r^2$ (the whole area). A slice is just that fraction of the whole.

Example: a 30 cm pizza (radius 15 cm) cut into 8 slices. Each slice angle is $\frac{2\pi}{8} = \frac{\pi}{4}$:

$$s = 15 \times \frac{\pi}{4} \approx 11.8 \text{ cm}$$ ← *crust per slice*

$$A = \tfrac{1}{2} \times 15^2 \times \frac{\pi}{4} \approx 88.4 \text{ cm}^2$$ ← cheese per slice

The same in degrees: radius 10 cm, angle $72°$ (a fifth of a turn):

$$s = \frac{72}{360} \times 2\pi \times 10 = 4\pi \approx 12.57 \text{ cm}$$ ← fraction × circumference

$$A = \frac{72}{360} \times \pi \times 10^2 = 20\pi \approx 62.83 \text{ cm}^2$$ ← fraction × area

PROCEDURE: finding r or θ from an arc or a sector

1. Write down what you know: r, θ, s or A. Decide degrees or radians.
2. Write the matching formula from the table with the known numbers put in.
3. Rearrange for the unknown (Chapter 14). For r from an area you get r^2 first: take the square root at the end.
4. Check by putting your answer back into the formula.

Find r: a sector has angle 1.5 rad and area 48 cm². Find the radius.

$$48 = \tfrac{1}{2} r^2 (1.5) = 0.75\, r^2 \;\Rightarrow\; r^2 = 64,\; r = 8 \text{ cm}$$

÷ 0.75, then √

Find θ: an arc of 14 cm on a circle of radius 8 cm. Find the angle in radians and degrees.

$$14 = 8\theta \;\Rightarrow\; \theta = 1.75 \text{ rad} = 1.75 \times \tfrac{180}{\pi} = 100.3°$$

$s = r\theta$

Find r in degrees: a $40°$ sector has an arc of 4π cm. Find r.

$$\frac{40}{360} \times 2\pi r = 4\pi \;\Rightarrow\; \frac{2\pi r}{9} = 4\pi \;\Rightarrow\; r = 18 \text{ cm}$$

✗ Common mistakes

✗ $s = r\theta = 5 \times 60 = 300$ ✓ $s = 5 \times \frac{\pi}{3} = 5.24$

$s = r\theta$ needs θ in radians: convert 60° first

Rules in this chapter

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Full turn	$360^\circ = 2\pi \text{ rad}$ half turn $180^\circ = \pi$; right angle $90^\circ = \frac{\pi}{2}$
Angle types	acute $< 90^\circ$; obtuse 90°–180°; reflex $> 180^\circ$
Clock hands	minute hand 6°/min; hour hand 30°/hour + 0.5°/min
Direction	from the positive x-axis; anticlockwise $+$, clockwise $-$
Degrees → radians	$\theta_{\text{rad}} = \theta_{\text{deg}} \times \dfrac{\pi}{180}$
Radians → degrees	$\theta_{\text{deg}} = \theta_{\text{rad}} \times \dfrac{180}{\pi}$
Reduce an angle	add or subtract 360° (2π) until it is in range
Arc length	$s = r\theta \quad \text{or} \quad s = \dfrac{\theta}{360} \times 2\pi r$ first with θ in radians, second in degrees
Sector area	$A = \frac{1}{2}r^2\theta \quad \text{or} \quad A = \dfrac{\theta}{360} \times \pi r^2$ first with θ in radians, second in degrees
Calculator	° → DEG mode; rad, π or no unit → RAD mode

∠ Your turn

Answers in grey.

Degrees

1. Angle between clock hands at 4:00
2. … at 9:30
3. Is 170° acute, obtuse or reflex?
4. Describe −90° as a turn

1. 120° 2. 105° (hour hand at 285°, minute at 180°) 3. obtuse 4. a quarter turn clockwise

Converting

5. 120° in radians
6. π/6 in degrees
7. 2.5 rad in degrees (1 d.p.)
8. 225° in radians

5. 2π/3 6. 30° 7. 143.2° 8. 5π/4

Arcs

9. Arc length: r = 10 cm, θ = 1.2 rad
10. Sector area: r = 4 m, θ = π/2
11. A slice has a crust of 6 cm on a 12 cm-radius pizza. Its angle in radians?

9. 12 cm 10. $4\pi \approx 12.6$ m² 11. 6 ÷ 12 = 0.5 rad

Find the unknown

12. A sector has angle 1.5 rad and area 27 cm². Find the radius.
13. An arc is 15 cm long on a circle of radius 6 cm. Find the angle in radians and in degrees.
14. A 40° sector has an arc of 6π cm. Find the radius.
15. A sector of radius 9 cm has area 27π cm². Find its angle in degrees.
16. Write 750° as an angle from 0° to 360°, and −5π/3 as an angle from 0 to 2π.

12. $r = 6$ cm ($\frac{1}{2}r^2 \times 1.5 = 27$, so $r^2 = 36$) 13. $\theta = 15 \div 6 = 2.5$ rad ≈ 143.2° 14. $r = 27$ cm ($(40/360) \times 2\pi r = 6\pi$, so $2\pi r/9 = 6\pi$) 15. 120° ($(\theta/360) \times 81\pi = 27\pi$, so $\theta/360 = 1/3$) 16. 750° − 720° = 30°; $-5\pi/3 + 2\pi = \pi/3$